CHAPTER TWO: LITERATURE REVIEW

Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference

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2.1 Introduction

This chapter reviews the existing body of literature relevant to the proposed study on predicting maternal health risk among pregnant women in Kenya using machine learning and Bayesian inference. The chapter is organised into three main sections. The first section presents a general literature review that contextualises the study within the broader landscape of maternal health, digital health innovation, and machine learning in healthcare. The second section provides an empirical literature review structured around the three research objectives of this study — namely the development of a machine learning classification model, the integration of Bayesian inference for probabilistic risk quantification, and the deployment of an AI-powered decision support tool for Community Health Workers (CHWs). The chapter concludes with a summary that identifies the critical research gap this study seeks to fill and positions the proposed work within current scholarly discourse.

The literature reviewed spans peer-reviewed journal articles, conference papers, and industry reports published primarily between 2023 and 2026, with emphasis on works from 2025 and 2026 to ensure currency and relevance. Special attention has been given to emerging deep learning architectures for tabular data, recent Bayesian modelling frameworks applied to health outcomes, and studies on explainable AI in clinical and community health worker contexts. Where Kenya-specific or Sub-Saharan Africa-specific literature was limited, closely related works from comparable low- and middle-income country (LMIC) settings have been included to establish methodological precedent.

2.2 General Literature Review

2.2.1 Maternal Mortality as a Global and Kenyan Public Health Crisis

Maternal mortality remains one of the most persistent indicators of health system failure globally. The World Health Organization (2023) estimates the global maternal mortality ratio at 223 deaths per 100,000 live births, with Sub-Saharan Africa bearing a disproportionate burden. Kenya's maternal mortality ratio stands at approximately 355 deaths per 100,000 live births, significantly above the global average and far from the Sustainable Development Goal 3.1 target of fewer than 70 deaths per 100,000 live births by 2030 (KDHS, 2023). The Kenya Demographic and Health Survey 2022 further reveals that only 66% of births in Kenya are attended by a skilled health professional, with stark disparities between counties — ranging from near-universal skilled

attendance in Nairobi to below 30% in northern counties such as Mandera and Turkana (KNBS & ICF, 2023).

The leading direct causes of maternal death in Kenya mirror global patterns: haemorrhage, hypertensive disorders of pregnancy, and sepsis account for the majority of deaths, compounded by indirect causes including anaemia and HIV (Say et al., 2014). Critically, the majority of these deaths are preventable with timely risk identification and appropriate referral. Opoku et al. (2025), in a scoping review protocol on risk prediction models for maternal and newborn outcomes in low- and middle-income countries published in PLOS ONE, concluded that the absence of validated, context-specific risk prediction tools is a primary contributor to preventable maternal deaths across LMICs, underscoring the urgent need for data-driven interventions tailored to national demographic profiles.

2.2.2 Kenya's Digital Health Landscape and Community Health Worker System

Kenya has made significant investments in digital health infrastructure over the past decade. The Government of Kenya passed four landmark health laws in 2023 — the Social Health Insurance Act, the Digital Health Act, the Primary Health Care Act, and the Facility Improvement Financing Act — signalling a policy commitment to digitally enabled universal health coverage (Kapsule, 2026). The Social Health Authority (SHA), which replaced the National Health Insurance Fund (NHIF) in October 2024, had enrolled over 27 million Kenyans by late 2025, positioning digital health identification as a cornerstone of the health system.

At the community level, Kenya has scaled its electronic Community Health Information System (eCHIS) nationally since 2020, a platform used by over 100,000 Community Health Promoters (CHPs) to report on household health services (PATH, 2025). In 2025, Cabinet Secretary Aden Duale reaffirmed the government's commitment to digitising community health systems and integrating machine learning capabilities into these platforms, explicitly stating that 'digital innovation is a critical enabler of Universal Health Coverage' (Ethical Business Africa, 2026). Despite this infrastructure investment, the eCHIS platform currently lacks any predictive risk scoring module — CHPs continue to rely on paper-based or protocol-driven assessments without access to data-driven individual risk classification tools. Ombech et al. (2025), evaluating a digital health system for maternal care in northern Kenya published in Frontiers in Digital Health, found that digital health tools significantly improved antenatal care uptake, but noted the absence of predictive analytics as a critical gap limiting proactive case management.

2.2.3 Machine Learning in Healthcare — Overview and Emerging Architectures

The application of machine learning in healthcare has grown substantially, with classification models increasingly used for disease diagnosis, risk stratification, and clinical decision support. Vasudevan et al. (2025), in a comprehensive scoping review of ML models for predicting maternal morbidity and mortality from electronic medical records published in the Journal of Medical Internet Research, identified 39 peer-reviewed studies demonstrating the feasibility and accuracy of ML approaches across a range of maternal outcomes. However, the review noted a critical limitation:

the vast majority of studies relied on hospital-based electronic health records (EHRs), making their models inapplicable in community or field settings where such data is unavailable.

While classical ensemble methods such as Random Forest and XGBoost have dominated health prediction literature, recent years have seen the emergence of sophisticated deep learning architectures specifically designed for tabular (structured) data — the format most relevant to health survey datasets like the KDHS. Doshi et al. (2025), comparing TabNet and TabTransformer against traditional ML methods for toxicity prediction in the Journal of Applied Toxicology, found that these novel architectures consistently outperformed classical classifiers on AUC-ROC and F1 metrics, particularly under class imbalance conditions. Sumon et al. (2025), proposing CardioTabNet — a hybrid TabTransformer architecture for heart disease prediction — published in Health Information Science and Systems, demonstrated that attention-based mechanisms applied to clinical tabular data capture complex non-linear feature interactions that tree-based models fail to model. Yasmeen et al. (2025), in Frontiers in Physiology, proposed a hybrid TabNet-stacked ensemble for heart disease prediction, reporting superior sensitivity compared to single-model approaches. These findings collectively support the adoption of TabNet and TabTransformer as novel, state-of-the-art modelling alternatives in the proposed study.

> *Research Direction: This study will evaluate TabNet and TabTransformer alongside classical ensemble methods (Random Forest, XGBoost, CatBoost) to identify the best-performing architecture for maternal risk classification on the KDHS 2022 tabular dataset — a comparison not yet attempted in the maternal health literature.*

2.2.4 Explainable AI (XAI) in Clinical Decision Support

A central challenge in deploying ML models in healthcare is the 'black box' problem — models that produce accurate predictions without providing interpretable reasoning fail to gain clinical trust. Explainable AI (XAI) has emerged as a critical field addressing this limitation. Abbas et al. (2025), in a meta-analysis of XAI in clinical decision support systems published in Healthcare, synthesised findings from 62 peer-reviewed studies and concluded that model-agnostic explainability methods — particularly SHAP (SHapley Additive exPlanations) and LIME — are the most widely adopted and clinically useful techniques for structured tabular data. Noor et al. (2025), in a comprehensive review published in WIREs Data Mining and Knowledge Discovery, identified the lack of standardised XAI evaluation metrics and insufficient CHW-oriented XAI design as two of the most critical gaps in the field.

A particularly relevant study by Mamun et al. (2025), published in Engineering Reports (Wiley), developed an explainable ML framework for maternal health risk detection achieving 99.51% accuracy using Bagging with PCA, incorporating SHAP and LIME for transparent model explanations. Their framework represents the current state of the art in explainable maternal health ML, though it was developed on a general dataset and does not address the KDHS context or CHW deployment constraints. Okolo et al. (2024), studying perceptions of XAI among CHWs in rural India, found that while CHWs initially struggled to interpret AI explanations, they strongly preferred AI tools that included explanations over those that did not — underscoring the importance of

designing plain-language explanation interfaces for non-technical frontline health workers, which the proposed Claude AI integration directly addresses.

2.3 Empirical Literature Review

This section reviews empirical studies directly related to each of the three research objectives of this study. For each objective, recent studies (2024–2026) are critically examined to identify methodological approaches, key findings, limitations, and research gaps that the proposed study addresses.

2.3.1 Objective 1 — Build a Machine Learning Classification Model for Individual Maternal Risk

2.3.1.1 ML Models Applied to Maternal Health Risk Prediction

Khadidos et al. (2025), published in Healthcare (Vol. 13, No. 5), applied boosting-based machine learning models to high-risk pregnancy prediction, finding that gradient-boosted models outperformed logistic regression on sensitivity metrics critical for healthcare applications. Their study used demographic and clinical variables comparable to those available in KDHS, validating the feature relevance of variables such as maternal age, parity, and antenatal care visit frequency for risk classification. However, their dataset was hospital-derived and excluded community-level wealth and socioeconomic indicators that are central to the KDHS-based approach proposed in this study.

Tzimourta et al. (2025), in Procedia Computer Science (Vol. 240), benchmarked multiple ML algorithms for predicting maternal health risk factors, reporting that ensemble models consistently outperformed single classifiers across precision, recall, and AUC metrics. Sarker et al. (2025), in Healthcare Analytics (Vol. 5), proposed a PCA-enhanced XGBoost model with SMOTE-ENN for maternal health risk prediction, achieving strong performance on imbalanced datasets — directly relevant to the class imbalance expected in the KDHS target variable. Their combination of dimensionality reduction and oversampling informs the preprocessing pipeline of the proposed study.

The most methodologically advanced recent work is the Quad-Ensemble Machine Learning framework (QEML-MHRC) proposed in Scientific Reports (2024), which combined Decision Tree, Random Forest, Gradient Boosting, and other base learners under a quad-ensemble architecture. Their framework achieved superior accuracy and robustness over single models, and directly supports the proposed study's adoption of a stacking ensemble approach. Crucially, no existing study has applied these ensemble or novel deep learning architectures to KDHS 2022 data at the individual level for Community Health Worker deployment in Kenya.

2.3.1.2 Novel Model Recommendation for This Study

> *Based on the literature, this study recommends adopting a three-tier modelling approach: (1) TabNet as the primary novel deep learning model — its sequential attention mechanism provides built-in feature selection interpretable to non-technical reviewers; (2) TabTransformer as the secondary novel model — its self-attention mechanism on categorical features is particularly suited to the predominantly categorical KDHS variables; and (3) a Stacking Ensemble combining the outputs of TabNet, XGBoost, and CatBoost under a Logistic Regression meta-learner. This architecture has not been applied to maternal health risk prediction in any reviewed study, constituting a methodological contribution of this project.*

2.3.2 Objective 2 — Bayesian Inference for Probabilistic Risk Quantification

The application of Bayesian inference to maternal health risk modelling has gained significant traction in 2025. Alhusseini et al. (2025), published in Taylor & Francis Online, applied Bayesian Accelerated Failure Time modelling with Elastic Net regularisation to a dataset of 549 maternal cases, demonstrating that Bayesian inference with regularisation effectively addresses data sparsity and rare events in maternal mortality data. Their study identified fewer than four antenatal care visits and young maternal age as significant risk factors — precisely the variables available in the KDHS dataset — validating the clinical alignment of the proposed study's feature set.

Sheridan et al. (2025), published in Computers in Biology and Medicine (Article 109975), proposed a Bayesian Network model of pregnancy outcomes for England and Wales, demonstrating that Bayesian networks produce more reliable and clinically interpretable risk estimates than conventional regression or ML classifiers for pregnancy outcomes. Their framework draws on publicly available population statistics — methodologically analogous to the KDHS data use proposed in this study — and validates the use of Bayesian models for national-level health survey data. The proposed study extends their approach by integrating Bayesian inference as an uncertainty quantification layer on top of an ML classification pipeline, a hybrid architecture not yet applied to Kenyan maternal health data.

A Bayesian hierarchical logistic regression approach to preterm birth risk by Bodnar et al. (in a related study) further demonstrated that posterior probability outputs with credible intervals provide substantially more actionable information for clinical decision-making than point predictions alone. The proposed study operationalises this finding by using PyMC to generate posterior probability distributions for each individual risk prediction, enabling the final app to display outputs such as 'High Risk — 84% probability (95% CI: 76%–91%)' rather than a binary classification.

> *Gap Identified: No study has combined a deep learning tabular model (TabNet/TabTransformer) with a Bayesian inference posterior layer to produce calibrated, uncertainty-quantified maternal risk scores on KDHS national survey data. This hybrid pipeline is the core methodological novelty of the proposed study.*

2.3.3 Objective 3 — AI-Powered Decision Support Tool for Community Health Workers

The deployment of AI-powered tools for community health workers has gained momentum globally, with studies increasingly demonstrating both feasibility and CHW preference for digitally assisted decision support. MHP Salud (2025) documented that CHW-supported AI data collection improved hospital readmission predictions by 5%, and noted ASHABot — an LLM-powered chatbot developed for CHWs in India — as a pioneering example of generative AI being deployed at the community health frontline. The proposed integration of the Claude API into a Streamlit application for Kenyan CHWs represents the first such deployment in the Kenyan maternal health context, building on and extending the ASHABot precedent.

Ombech et al. (2025), evaluating a community-hospital digital health system for maternal and newborn care in northern Kenya in Frontiers in Digital Health, found statistically significant improvements in antenatal care uptake and skilled delivery rates following the introduction of a digital health information system. Their study explicitly recommended the addition of predictive analytics and risk scoring functionality as the logical next evolution of Kenya's eCHIS platform — directly motivating the proposed study's deployment target. Similarly, a 2025 study evaluating community digital data linkage for antenatal care uptake in Western Kenya (Frontiers in Health Services) confirmed that eCHIS adoption is now widespread enough that a risk prediction layer would reach a significant proportion of Kenya's CHW workforce.

On the use of large language models (LLMs) in clinical explanation, Abbas et al. (2025) and Noor et al. (2025) both noted the emerging role of generative AI in translating ML outputs into plain-language clinical guidance, identifying it as one of the most promising frontiers in health AI. The proposed Claude AI integration directly operationalises this finding, using the Anthropic API to generate contextualised explanations of SHAP-derived risk drivers, care recommendations, and referral letters in both English and Swahili — a bilingual capability not present in any reviewed maternal health AI tool.

PSO-optimized TabTransformer for cervical cancer risk prediction (ScienceDirect, 2025) further noted that 'the rapid advancement of large language models has significantly impacted healthcare, enabling automated clinical reasoning, medical text summarisation, and knowledge extraction' — confirming that integrating LLMs with predictive models represents the current frontier of health AI research, and that the proposed architecture is aligned with the most recent scholarly direction.

2.4 Chapter Summary

This chapter has reviewed the theoretical and empirical literature underpinning the proposed study across three dimensions: the global and Kenyan maternal health crisis, the evolution of machine learning and explainable AI in healthcare, and the specific evidence base for each of the study's three research objectives.

The general literature review established that Kenya's maternal mortality ratio of 355 per 100,000 live births remains critically high and disproportionately affects women in rural and underserved counties. Kenya's digital health infrastructure — anchored by the eCHIS platform and over 100,000 CHPs — provides a ready deployment pathway for a predictive risk tool, yet no such ML-based module currently exists within Kenya's community health system. The literature further established that novel deep learning architectures for tabular data — particularly TabNet and TabTransformer — represent a significant advancement over classical ensemble methods, and that XAI techniques including SHAP and LIME are essential for clinical trust and CHW usability.

The empirical literature review, organised by research objective, confirmed that: (1) ensemble and deep learning models consistently outperform single classifiers for maternal risk prediction, with no existing study applying TabNet or TabTransformer to KDHS data; (2) Bayesian inference provides superior uncertainty quantification for maternal health outcomes compared to point-estimate ML models, with no hybrid Bayesian-deep learning pipeline yet applied to Kenyan maternal data; and (3) AI-powered CHW tools have demonstrated measurable health impact in comparable LMIC settings, with generative AI identified as an emerging frontier for translating ML outputs into actionable clinical guidance.

Collectively, these findings confirm that the proposed study addresses a genuine, documented, and unresolved research gap in Kenyan health informatics. The combination of TabNet/TabTransformer modelling, Bayesian uncertainty quantification, SHAP explainability, and Claude AI-powered CHW guidance — applied to KDHS 2022 individual-level data — constitutes an original methodological and applied contribution to the field. The following chapter will detail the research methodology that operationalises these findings into a coherent and implementable research design.

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